

Navneet Kumar Dubey

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Technical Skills

Languages: Kotlin, Java, TypeScript, C

Frameworks & Libraries: Jetpack Compose, WebRTC, Node.js (Express), WebSockets, React

Databases & APIs: PostgreSQL, Room (SQLite), Redis, Geolocation APIs, Mapbox/Google Maps API

Tools & Platforms: Git/GitHub, MinIO, Firebase, Google Cloud Console, AI Object Detection Modules

Education

B.M.S. College of Engineering

B.E. in Computer Science and Engineering – CGPA: 8.5/10

2024 – 2028

Bengaluru, India

Projects

Real-Time Telehealth App – Kotlin, Jetpack Compose, Room, Node.js, Redis, MinIO, PostgreSQL, WebSockets, V

- Engineered a high-performance, real-time application featuring production-grade video streaming and state architecture.
- Implemented a reliable **WebRTC video streaming** protocol combined with a custom-built **WebSocket signaling server** to establish seamless, low-latency communication.
- Designed a robust data layer integrating local persistence via Android **Room (SQLite)** with server-side **PostgreSQL** relational architectures, **MinIO** object storage for media, and **Redis** for high-speed caching.

AI Civics Mapper (Athernex Hackathon) – React, Node.js, AI Object Detection, Map Integrations

- Co-developed an automated civic monitoring web platform during the Athernex Hackathon to detect infrastructural hazards.
- Integrated an **AI computer vision modeling** pipeline capable of parsing media frames to automatically detect potholes and garbage.
- Programmed real-time interactive mapping elements that dynamically plot identified markers along with an automated calculated **severity score** for city maintenance scheduling.

Mood-Driven Social Travel Planner – TypeScript, React, Geolocation APIs, WebSockets

- Designed and built a full-stack automated single-day itinerary generator within a rapid **4-hour hackathon environment**.
- Developed an algorithmic parser that transforms simple natural language text user inputs into structured 1-day travel timelines detailing route stops, mapping configurations, and optimized timings.
- Engineered live social interaction modules utilizing persistent network channels, allowing users to create itineraries and invite peer networks to join their active groups synchronously.

Terminal Chess Engine – TypeScript

- Developed a self-contained interactive chess game running within the command-line interface consisting of over **2,400+ lines of native TypeScript**.
- Programmed comprehensive rule verification using explicit **brute-force decision structures** to validate multi-tier piece movements and complex board states.

Experience

Full-Stack Developer – UTSAV 2025

- Collaborated within a student engineering team to design and deploy frontend features and backend integrations supporting the college's annual techno-cultural festival website.

Achievements

- **LeetCode Problem Solving:** Solved over **170+ algorithmic problems**, maintaining a strong foundation across core Data Structures and Algorithms (DSA).
- **Athernex Hackathon:** Successfully shortlisted as a competitive participant representing B.M.S. College of Engineering in a rapid technical development format.